Book 1 — Lecture 1 — Interlude 1 — Exercise 5

Orthogonal Vector Pair

Prompt. Determine which pair of vectors are orthogonal: $(1, 1, 1)$, $(2, -1, 3)$, $(3, 1, 0)$, and $(-3, 0, 2)$.

Method

Two vectors are orthogonal exactly when their dot product is zero[1].

Pairwise dot products

$$\begin{aligned}
(1, 1, 1) \cdot (2, -1, 3) &= 4, \\
(1, 1, 1) \cdot (3, 1, 0) &= 4, \\
(1, 1, 1) \cdot (-3, 0, 2) &= -1, \\
(2, -1, 3) \cdot (3, 1, 0) &= 5, \\
(2, -1, 3) \cdot (-3, 0, 2) &= 0, \\
(3, 1, 0) \cdot (-3, 0, 2) &= -9.
\end{aligned}$$

Result. The only orthogonal pair is $(2, -1, 3)$ and $(-3, 0, 2)$.

Example. Check:

$$(2, -1, 3) \cdot (-3, 0, 2) = 2(-3) + (-1)0 + 3(2) = -6 + 0 + 6 = 0.$$

Since the dot product is zero, these vectors are orthogonal.

[1] For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, $\mathbf{u} \cdot \mathbf{v} = 0$ means the angle between them is $90°$.